

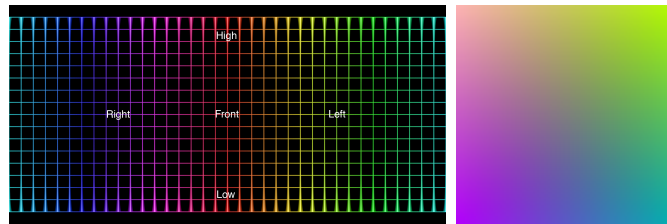
CSSE490 Textures Writeup

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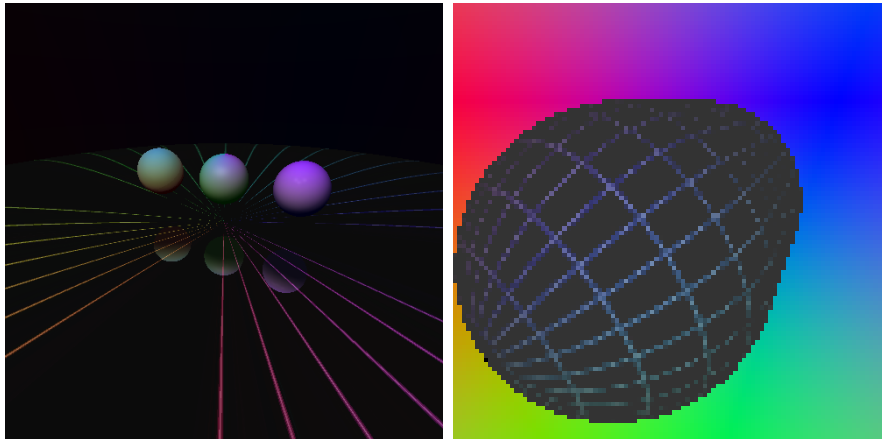
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Results

Our goal was to generate two scenes, spheres and capsule. We were provided with the following textures:



We succeeded in this, and were able to generate capsule & spheres. Capsule had the grid texture, and spheres had the grid texture for the bottom sphere, while the upper three spheres had the gradient. Our results are below:



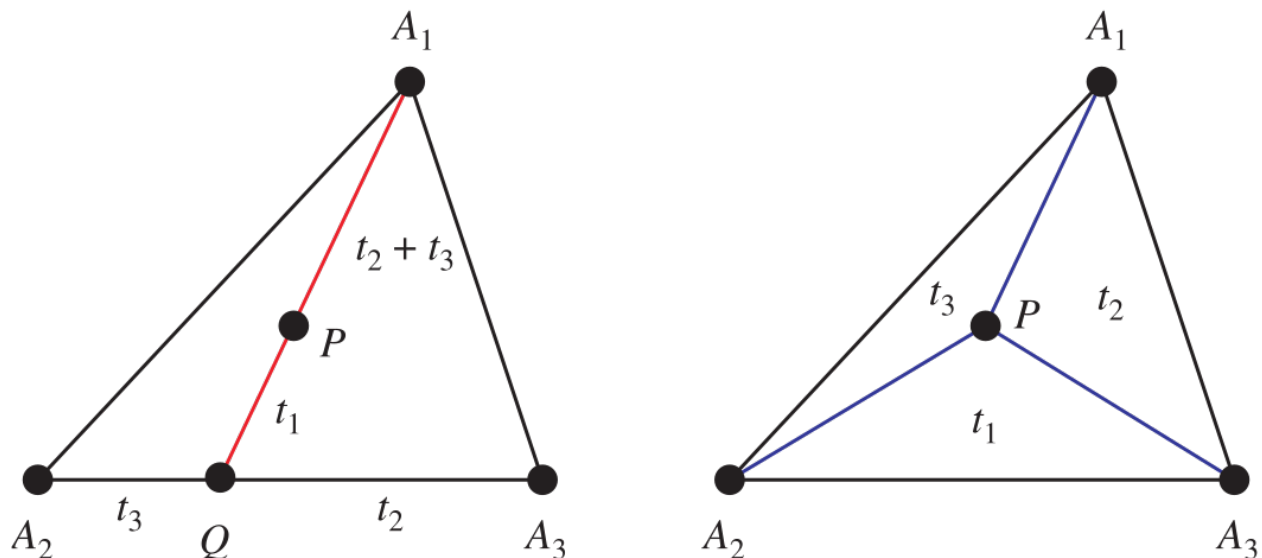
We can still see the material effects clearly in spheres, as each of the small spheres has slightly different coloration due to the material properties, along with the reflectivity of the big sphere still being present.

Methods

Textures rely on the existence of a local coordinate system for every object.

Triangles

For triangles, this system is the barycentric coordinate system:



<https://mathworld.wolfram.com/BarycentricCoordinates.html>

When a ray hits a triangle, the point at which it hits is converted to barycentric coordinates. The triangle is first divided into three subtriangles which are formed by drawing lines between each of the triangles three vertices and the hit point. We find the area of each of these triangles. Then we divide the subtriangle area by the area of the entire triangle. Doing this for each of the three triangles gives us the barycentric coordinates for the hitpoint. To find where on the texture the hitpoint corresponds to, we take the triangle in UV space and the barycentric coordinates of the hitpoint to find the corresponding hitpoint in UV space with the following calculation:

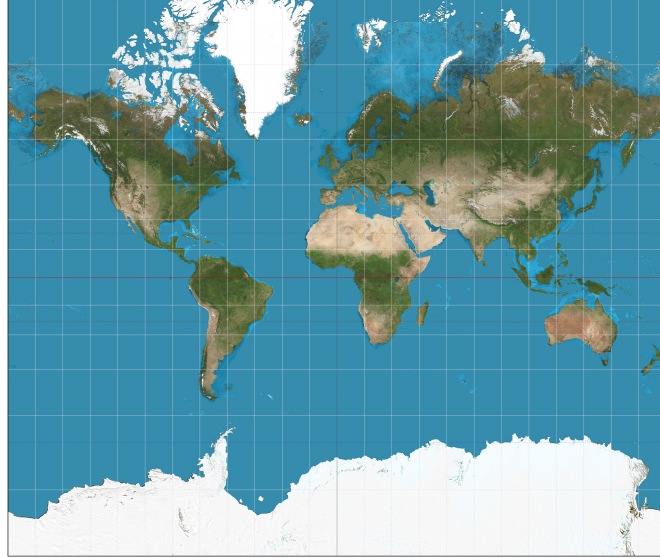
$$u = A_1.u * b_1 + A_2.u * b_2 + A_3.u * b_3$$

$$v = A_1.v * b_1 + A_2.v * b_2 + A_3.v * b_3$$

Where $A_1.u$ is the u coordinate of the first vertex of the triangle in UV space, b_1 is the first component of the barycentric coordinate of the hitpoint, and so on. This gives us the UV coordinates of the hitpoint.

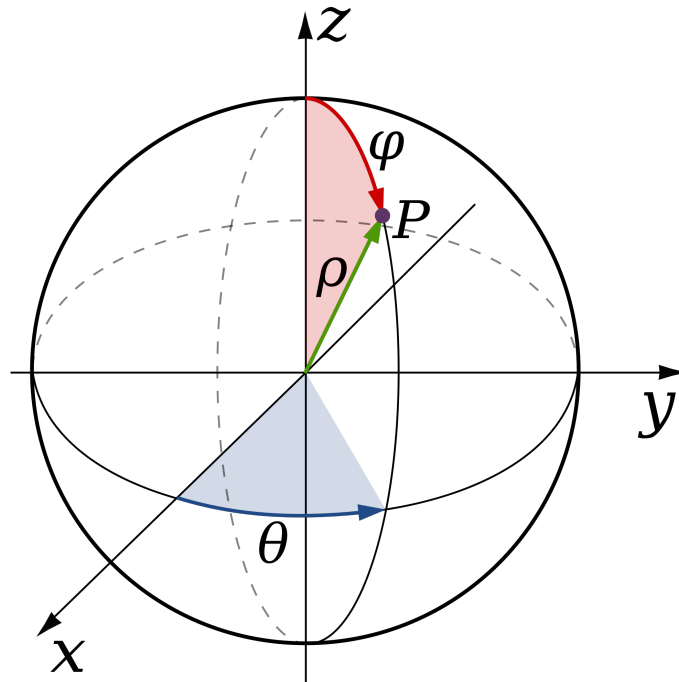
Spheres

Spheres cannot be easily mapped to a flat, square, two dimensional coordinate system. There are various projections that can create useful mappings between the two, but they are all flawed. Famously, the Mercator projection results in a distortion of objects higher up on a sphere, making them appear “taller”.



https://en.wikipedia.org/wiki/File:Mercator_projection_SW.jpg

The projection we used was simply spherical coordinates:



Polar coordinates have three coordinate values, θ , φ , and ρ . As we are describing spheres, ρ is constant. Therefore, we are left with two coordinates, θ and φ . We have $u = \frac{\theta}{2\pi}$. Notably, φ increases as we travel down the sphere, so our $v = 1 - \frac{\varphi}{\pi}$, such that the top of the sphere will have $v = 1$, and the bottom will have $v = 0$. These values must be calculated using inverse sines and cosines.